

Mitochondrial and nuclear DNA damage induced by curcumin in human hepatoma G2 cells.

Curcumin is extensively used as a spice and pigment and has anticarcinogenic effects that could be linked to its antioxidant properties. However, some studies suggest that this natural compound possesses both pro- and antioxidative effects. In this study, we found that curcumin induced DNA damage to both the mitochondrial and nuclear genomes in human hepatoma G2 cells. Using quantitative polymerase chain reaction and immunocytochemistry staining of 8-hydroxydeoxyguanosine, we demonstrated that curcumin induced dose-dependent damage in both the mitochondrial and nuclear genomes and that the mitochondrial damage was more extensive. Nuclear DNA fragments were also evident in comet assays. The mechanism underlies the elevated level of reactive oxygen species and lipid peroxidation generated by curcumin. The lack of DNA damage at low doses suggested that low levels of curcumin does not induce DNA damage and may play an antioxidant role in carcinogenesis. But at high doses, we found that curcumin imposed oxidative stress and damaged DNA. These data reinforce the hypothesis that curcumin plays a conflicting dual role in carcinogenesis. Also, the extensive mitochondrial DNA damage might be an initial event triggering curcumin-induced cell death.